Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice

Bentley Oakes[1], Michalis Famelis[2], Houari Sahraoui[2]

[1]Polytechnique Montréal
[2]Université de Montréal
bentley.oakes@polymtl.ca





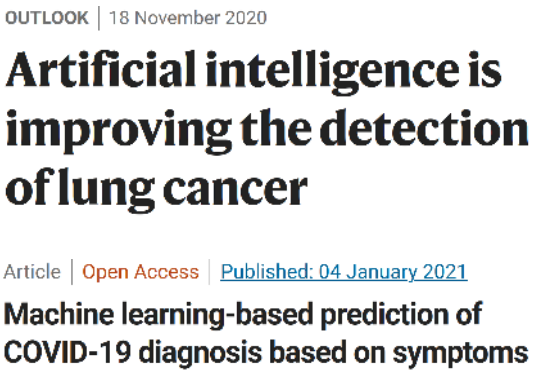
TOSEM Paper



- Subject of my second post-doc
- at GEODES SE lab, UdeM
- Submitted to TOSEM in 2022
- Published in 2024

- Looking for Master's candidate to continue topic
- Post-doc at McGill on topic
- Selected for ICSE journal-first presentation and poster

Domain experts want to use *machine learning* (ML):


OUTLOOK | 18 November 2020

Artificial intelligence is improving the detection of lung cancer

Article | Open Access | Published: 04 January 2021

Machine learning-based prediction of COVID-19 diagnosis based on symptoms

but ML requires significant knowledge

- Type of ML to use
- How to structure data for use
- Training, under/overfitting
- Programming library APIs
- Parameter setting

Motivation - Who's the User?

Imagine neuroscientist with research problem:

"With patient MRI data, how effective is machine learning at detecting depression?"

Assuming neuroscientist prefers to:

- Learn minimal amount of machine learning to solve problem
- Reason about neuroscience more than machine learning
- Utilise neuroscience knowledge to improve classification performance

Meaning of Domain-specific

From domain-specific language research, domain experts are more efficient when reasoning about familiar, domain-specific things.

For neuroscientist:

- Concepts neurons, activation, and brain damage
- Data formats like MRI images
- Data processing like loading MRI images, skull stripping, and motion correction

Here, domain-specific means
familiar only to domain experts, and not useful to others.

Research Aim

How can we (software engineers) help domain experts use ML?

Sub-questions:

- How can domain expert decisions be structured?
- How do available tools assist these decisions?
- How are domain experts using tools?
- How can domain-specific knowledge be utilised?
  - To create and improve solutions?
  - To improve machine learning performance?

Overview

1 Framework and Challenge Questions

2 Workflow Tools and Support

3 Real-World Examples

4 Conclusion and Next Steps

Outline

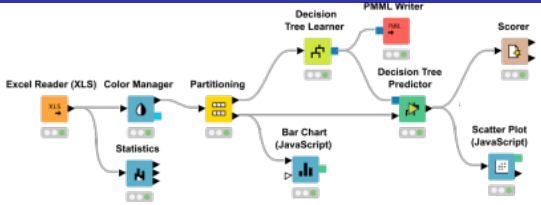
1 Framework and Challenge Questions

2 Workflow Tools and Support

3 Real-World Examples

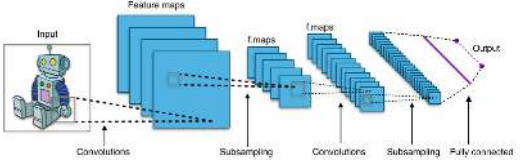
4 Conclusion and Next Steps

"Low-code" platforms like KNIME

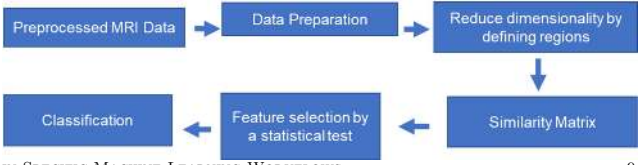


ML pipelines

ML models such as neural nets

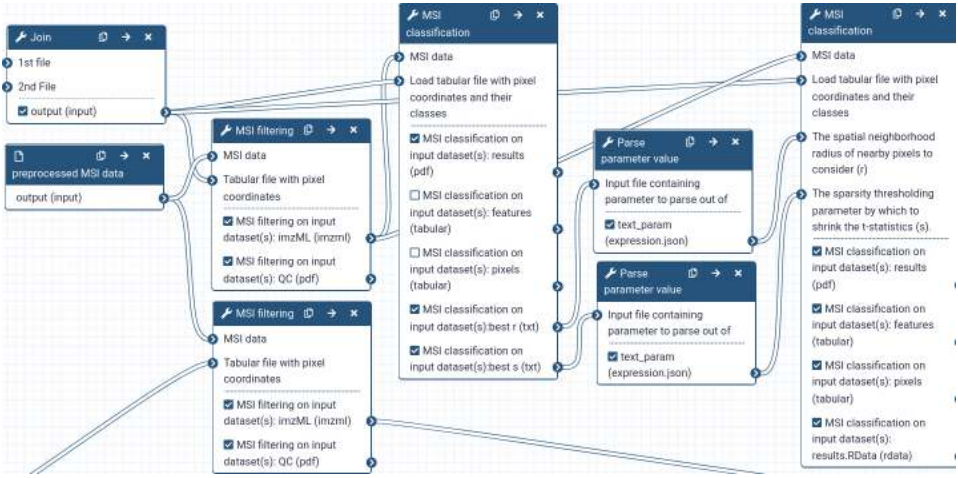


Scientific workflows



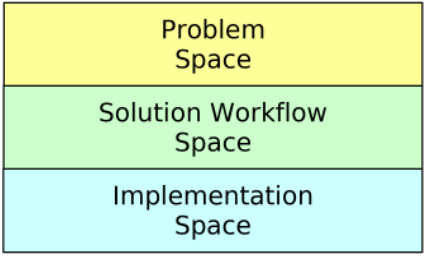
Computational Workflows

Directed graph of: Typed components, and acyclic or cyclic control/data flow



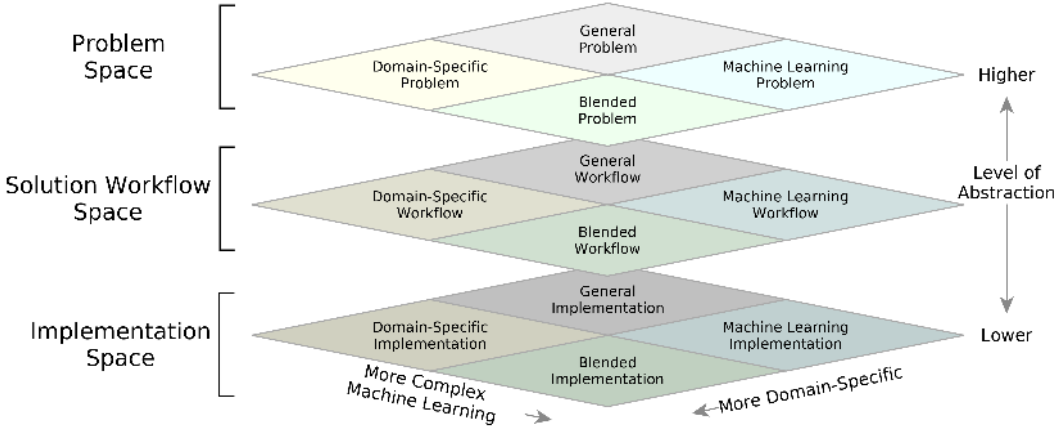
Benefits: FAIRness/provenance, abstraction, plugins for ML/DS

Three-Layer Framework



How can a domain expert start with a *domain-specific problem*,
efficiently find a *solution workflow*,
and then obtain an *implementation*?

Layers with Two Dimensions



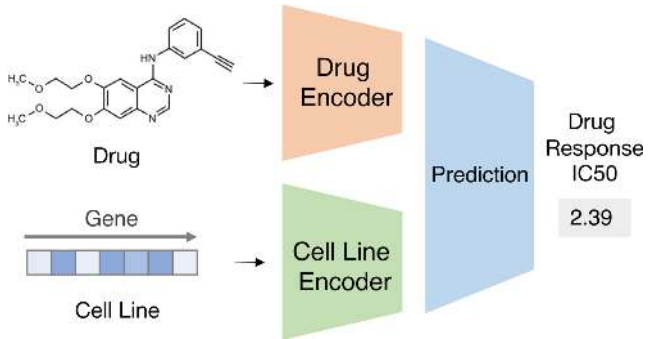
- How domain-specific the problem/workflow/implementation is
- Presence of machine learning concepts/components/API calls

Dimensions Example - Problem

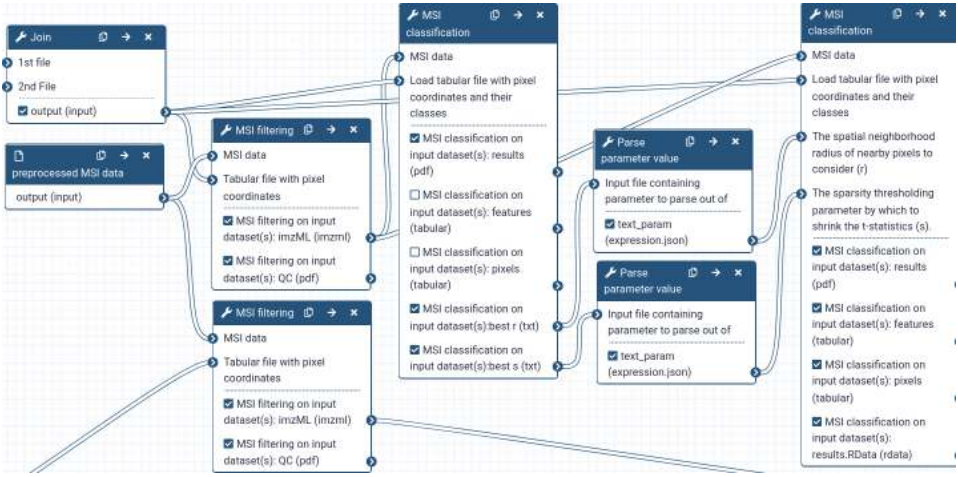
What does more domain-specific and more machine learning mean?

Domain-specific Problem: *Predict drug efficacy on tissue sample*
Machine Learning Problem: *Provide a numerical prediction for efficacy of graph structure versus linear string of characters*

Blended Problem:

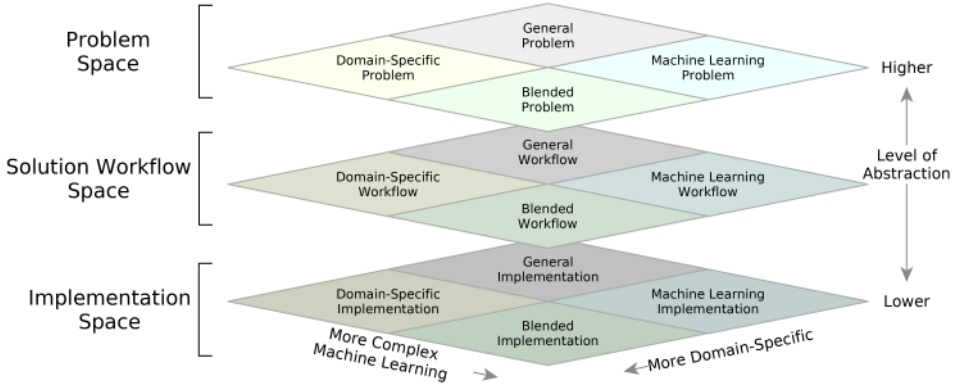


Dimensions Example - Domain-specific Workflow



- *Join* and *Parse* components are *general*
- MSI components are domain-specific, for mass spectrometry imaging

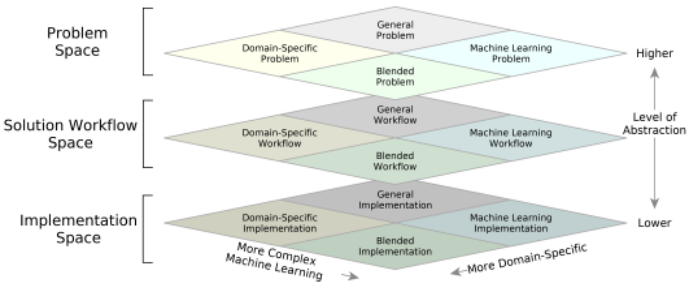
Transformations in Framework



- What does it mean to move along a dimension? e.g. How can a workflow increase in domain-specificness (DS) or involve more machine learning (ML)?
- What are the transformations moving from one layer to another?
- What regions are domain experts passing through?

Domain Expert Challenges

Transformations between regions are actions/choices for domain expert:



A) Adding ML concepts to DS problem

B) Obtaining workflow from problem

C) Making workflow more ML

D) Making workflow more DS

E) Obtaining implementation from workflow

F) Obtaining workflow from implementation

- Do current workflow tools support these?
- What choices are domain experts currently making?
- Gain insight into best practices available to help domain experts

Outline

1 Framework and Challenge Questions

2 Workflow Tools and Support

3 Real-World Examples

4 Conclusion and Next Steps

Tool Overview

Article is **not** a systematic review of workflow tools

`https://s.apache.org/existing-workflow-systems` lists
more than 300 workflow systems

Instead, chose tools:

- From various domains and used in publications
- To examine transformation possibilities
- To examine assistance for including domain-specific and machine learning knowledge

Next slides are brief overview of four tools

nipype - Textual Language Module

Python library providing concepts like *workflows*, *nodes*, and *neuroscience-specific tool interfaces*
(brain extraction, brain tissue segmentation)

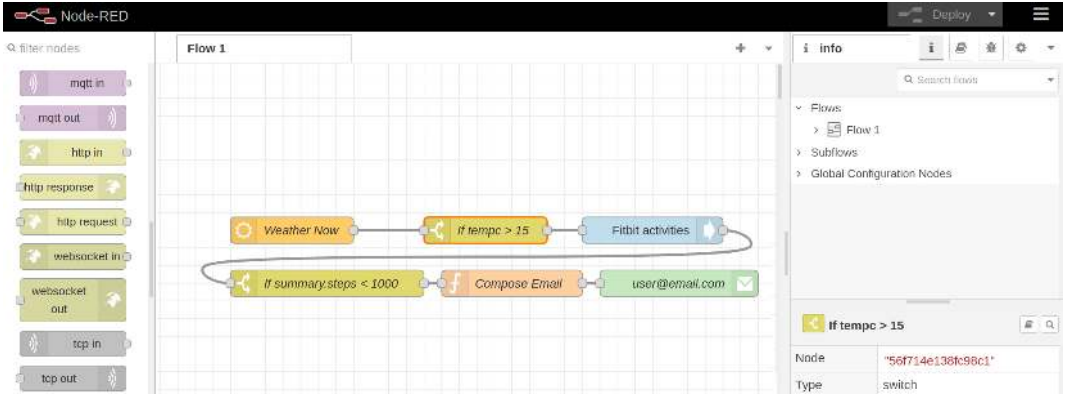


- Can build *workflow* out of existing *implementation*
- Standardisation of neuroscience-specific processing
  - fMRIPrep - standardised workflow for fMRI processing
  - *Workflow* and *implementation* are then domain-specific

Machine learning?

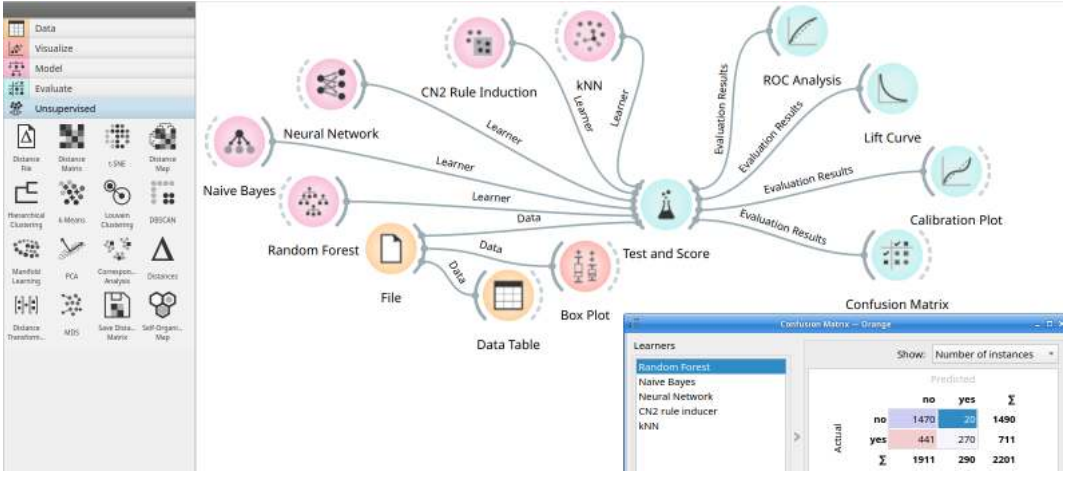
Not directly, but fMRIPrep often used to pre-process data

Node-RED - "Low-code" Graphical IoT/ML



- Palette approach for selecting components
- Plug-ins for devices/domains (including machine learning)

Orange - "Low-code" Graphical ML Tool



- Also plug-in approach
- Excellent for experimentation, but poor explanations

Galaxy - Graphical DS Platform

Web-based computational biology platform



- Offers DS support: datasets, component libraries, DS platform deployment
- ML component library
- Focus on provenance, stores all histories of component input/output
- Built-in component recommendation system

Transformation Support in Thirteen Tools

Question	Techniques	Supported
a) Mapping DS → ML	Expert mapping, ontologies, LLMs	○
b) Problem → Workflow	Workflow composition, examples, repos, libraries	●
c) Increasing workflow ML	Suggestions, experimentation	◐
d) Increasing workflow DS	Knowledge representation, feature extraction	◐
e) Workflow → Implementation	Containerisation, deployment, run within tool	●
f) Implementation → Workflow	Code mining, language integration	◔

- Strong support moving between framework layers
- Weak support for moving along DS or ML dimensions
- Automation mostly for question e), some b)

Outline

1 Framework and Challenge Questions

2 Workflow Tools and Support

3 Real-World Examples

4 Conclusion and Next Steps

Examples Overview

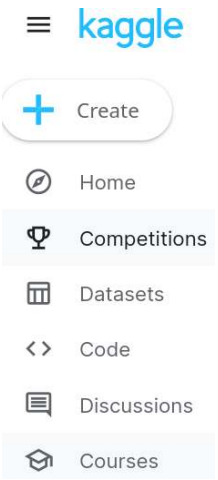
Selected six DS real-world examples utilising ML:

Tool	Domain	Repres.	Workflow Expression	Regions
PyTorch	Chemistry	Textual	Implicit	B-B-B
MATLAB	Geochemistry	Textual	Implicit	B-B-B
Kaggle	Marine biology	Textual	Implicit	ML-BML-BML
nipype	Neuroscience	Textual	Hybrid	B-DS-DS
Orange	Data mining	Graphical	Explicit	ML-ML-ML
Galaxy	Bioinformatics	Graphical	Explicit	DS-DS-DS

Ad-hoc selection for heterogenity, not systematic

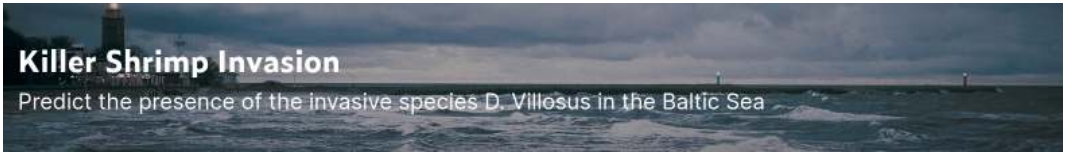
Three examples shown next

Kaggle Example - Background



- Data science website, hosts challenges by organisations
- Organisations provide context, problem, and data
- Participants build ML Jupyter notebooks, compete on best solution
- Dialogue during competition and after to share DS/ML knowledge

Kaggle Example - Killer Shrimp Challenge



Challenge is to:

- Take oceangraphic data (temperature, salinity, depth, etc.), and
- Predict the presence or not of invasive species



Problem: ML

- Input data is tabular
- Predicting one or zero in a certain column

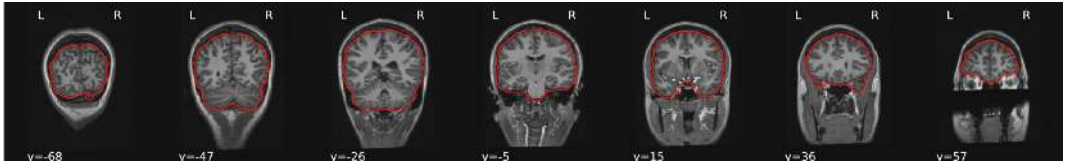
Kaggle Example - Workflow

Examined public Jupyter notebook of second place winner:

High-level Step	Step	Classification
Obtain Data	Load files	General
	Clean data	ML
Feature Adding	Add Water Density	DS
	Classify Wave Exposure	DS
	Add Polynomial Temp. Feature	ML
	Add Outlier Feature	ML
Train	Train Classifier	ML
Predict	Prediction	ML
Analysis	Write out	General
	Plot Feature Importance	ML

- *Workflow* is mostly *implicit*, but Jupyter notebook has explicit *cells*
- Was not expecting DS steps
- *Implementation* is somewhat *blended*, but mostly ML

nipype Example - MRI Depression



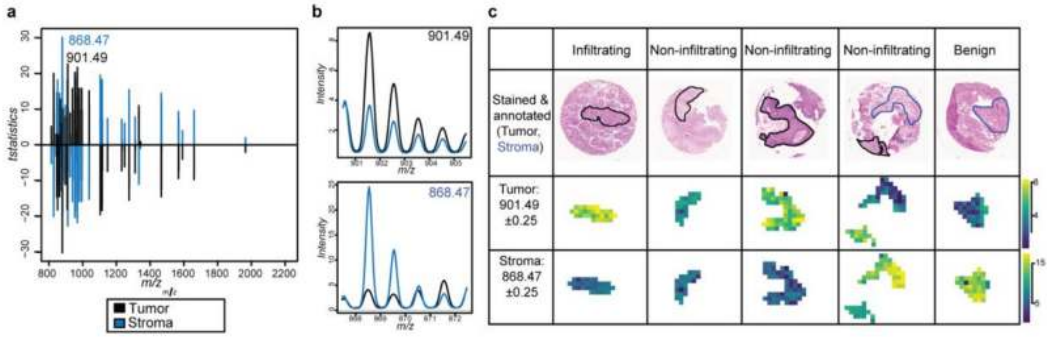
Problem: Blended - Classify depression based on MRI images

- Correlate MRI voxels
- Use correlations to classify depression w/ 14 classifiers

Workflow:
Hybrid - defined in Python, but fMRIPrep step defined in nipype
Overall, DS - most workflow components are DS.

High-level Step	Step	Classification
Pre-Process Data	Load data	General
	Convert to BIDS	DS
	fMRIPrep	DS
	FSL FEAT	DS
Defining Features	Data Preparation	DS
	Prep. Initial Features	DS
	Feature Selection	ML
Classification	Classification	ML

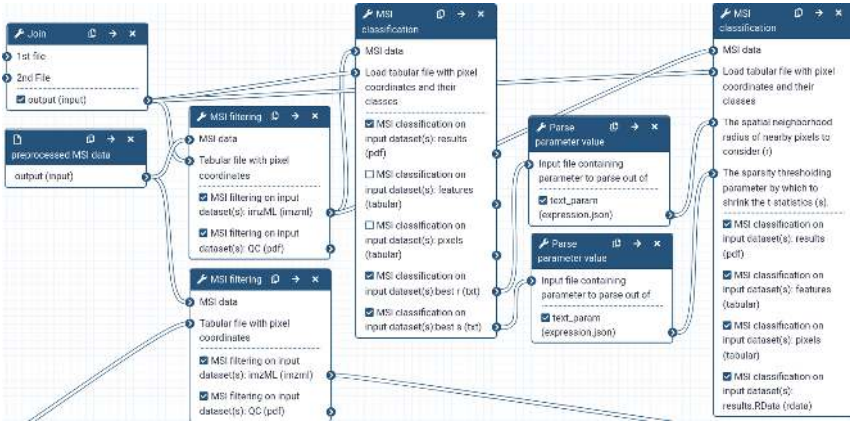
Galaxy Example - Tumour Problem



Problem - DS:

- *Mass spectrometry imaging* is done on slice of tissue
- Creates a one-dimensional spectrum for each "pixel" of the image.
- Based on this spectrum, classify the pixel as *infiltrating/non-infiltrating tumour* or *stroma* (connective tissue)

Galaxy Example - Workflow



Workflow: Explicit - Very DS

- Components previously created by authors
- Wrappers around DS implementations defined in R

Outline

1 Framework and Challenge Questions

2 Workflow Tools and Support

3 Real-World Examples

4 Conclusion and Next Steps

Contributions and Takeaways

- Three-layer framework with domain-specific and machine learning dimensions
- Defined transformations as challenge questions:
  - A) Adding ML concepts to DS problem
  - B) Obtaining workflow from problem
  - C) Making workflow more ML
  - D) Making workflow more DS
  - E) Obtaining implementation from workflow
  - F) Obtaining workflow from implementation
- Examined state-of-practice tools and real-world examples

- Wide range of techniques available to assist domain experts
- Best practices uneven across fields
- More structure and automation needed

Next Steps

- What software engineering techniques to introduce?
- How should domain expert be guided?
  - How to customise ML/tool advice per domain?
- How can domain expert be eased into ML, slowly providing options?
- How much automation to provide?
  - How to build this low-code tool?
  - AutoML/LLMs can auto-create workflows
  - How to use principles of *human-guided ML*?

Thank you!

Building Domain-Specific Machine Learning Workflows: A Conceptual Framework for the State-of-the-Practice

`https://dl.acm.org/doi/abs/10.1145/3638243`

Adding Domain-Specific Knowledge is Desirable?

> *"Don't worry about having domain knowledge to attempt a specific problem. The main thing we learned in this competition is that you don't necessarily need domain knowledge or industry [sic] to successfully tackle the data science challenge. Sometimes it even can be an advantage, as you go in blindly without many prior assumptions that might wrongly steer your exploratory analyses."*

https://medium.com/kaggle-blog/
from-football-newbies-to-nfl-data-champions-a-winners-interview-with-the-zoo-391793168714